Nagar YuwakShikshanSanstha's
YeshwantraoChavan College of Engineering
(An Autonomous Institution affiliated to RashtrasantTukadojiMaharaj Nagpur University)
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Department of Computer Technology

Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skills through emerging technologies.

Session 2025-2026

Vision: *To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration*	**Mission:** *To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skills through emerging technologies.*

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

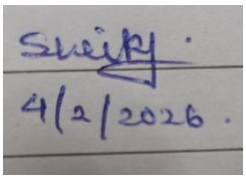
Keywords of POs:
Engineering knowledge,Problem analysis, Design/development of solutions,Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World,Ethics, Individual and Collaborative Team work,Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Name and Signature of Student and Date
(Signature and Date in Handwritten)
Sheikh Wasimuddin



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Session	2025-26 (EVEN)	Course Name	Competitive Coding Lab
Semester	6	Course Code	23IOT1605
Roll No	63	Name of Student	Sheikh Wasimuddin
Practical Number	CC Slot -2		

1. Write a Program to Check if a number is an Armstrong number or not.

```
#include <stdio.h>
int main() {
 int n, r, sum = 0, temp;
 printf("Enter number: ");
 scanf("%d", &n);
 temp = n;
 while (temp > 0) {
 r = temp % 10;
 sum = sum + (r * r * r);
 temp = temp / 10;
 }
 if (sum == n)
 printf("Armstrong Number");
 else
 printf("Not Armstrong Number");
 return 0;
}
```

main.c

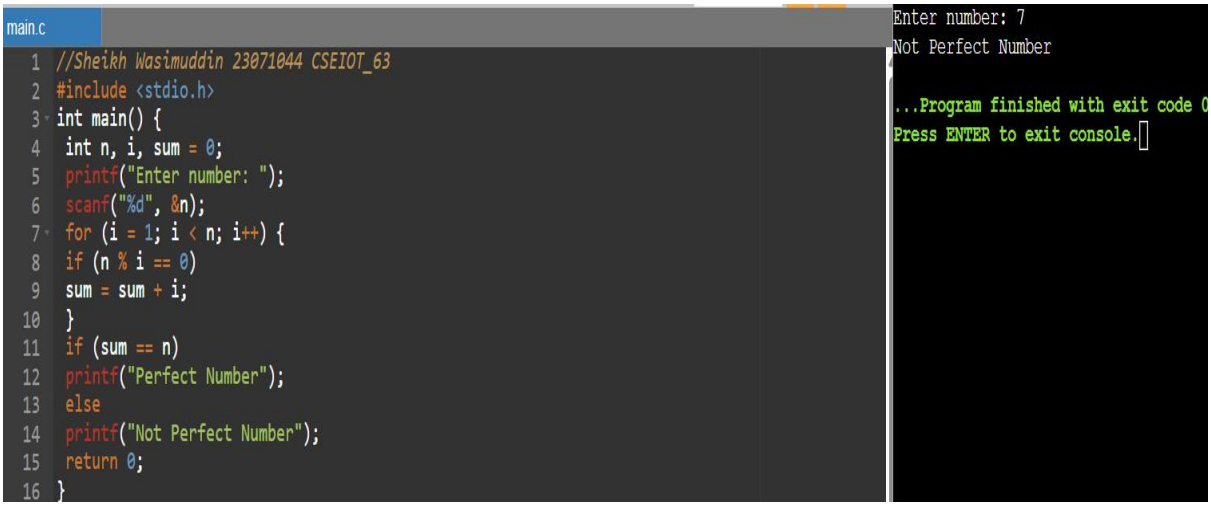
```c
//Sheikh Wasimuddin 23071044 CSEIOT_63
#include <stdio.h>
int main() {
 int n, r, sum = 0, temp;
 printf("Enter number: ");
 scanf("%d", &n);
 temp = n;
 while (temp > 0) {
 r = temp % 10;
 sum = sum + (r * r * r);
 temp = temp / 10;
 }
 if (sum == n)
 printf("Armstrong Number");
 else
 printf("Not Armstrong Number");
 return 0;
}
```

```
Enter number: 153
Armstrong Number

...Program finished with exit code 0
Press ENTER to exit console.
```

2. Write a Program to Check if a number is the perfect number or not.

```c
#include <stdio.h>
int main() {
 int n, i, sum = 0;
 printf("Enter number: ");
 scanf("%d", &n);
 for (i = 1; i < n; i++) {
 if (n % i == 0)
 sum = sum + i;
 }
 if (sum == n)
 printf("Perfect Number");
 else
 printf("Not Perfect Number");
 return 0;
}
```



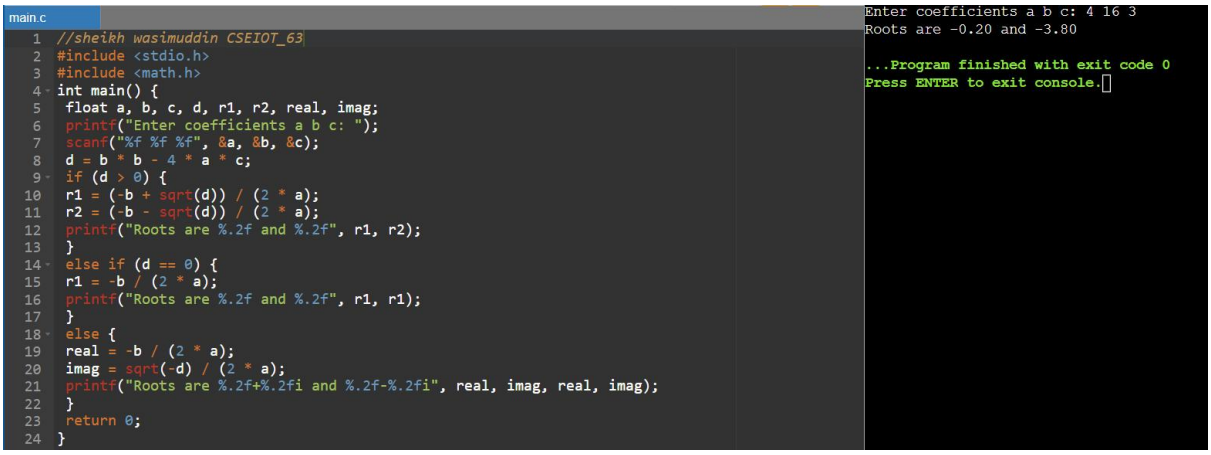
3. C Program to generate Fibonacci Series.

```
#include <stdio.h>
int main() {
 int n, i;
 int a = 0, b = 1, c;
 printf("Enter number of terms: ");
 scanf("%d", &n);
 printf("Fibonacci Series: ");
 for (i = 1; i <= n; i++) {
 printf("%d ", a);
 c = a + b;
 a = b;
 b = c;
 }
 return 0;
}
```

4. Write a program to find all the roots of a quadratic equation in C.

```
#include <stdio.h>
#include <math.h>
int main() {
 float a, b, c, d, r1, r2, real, imag;
 printf("Enter coefficients a b c: ");
 scanf("%f %f %f", &a, &b, &c);
 d = b * b - 4 * a * c;
 if (d > 0) {
 r1 = (-b + sqrt(d)) / (2 * a);
 r2 = (-b - sqrt(d)) / (2 * a);
 printf("Roots are %.2f and %.2f", r1, r2);
 }
 else if (d == 0) {
 r1 = -b / (2 * a);
 printf("Roots are %.2f and %.2f", r1, r1);
 }
 else {
 real = -b / (2 * a);
 imag = sqrt(-d) / (2 * a);
 printf("Roots are %.2f+%.2fi and %.2f-%.2fi", real, imag, real,
imag);
```

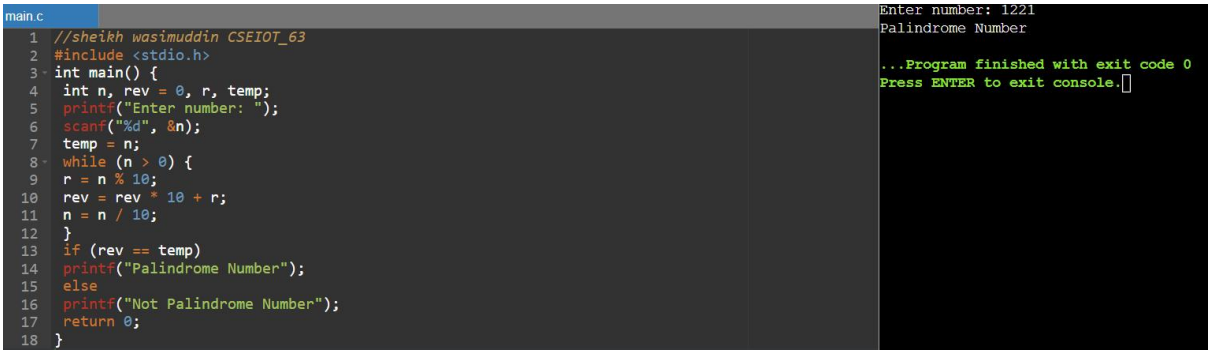
```
 }
 return 0;
}
```



5. Check whether a number is a palindrome.

```
#include <stdio.h>
int main() {
 int n, rev = 0, r, temp;
 printf("Enter number: ");
 scanf("%d", &n);
 temp = n;
 while (n > 0) {
 r = n % 10;
 rev = rev * 10 + r;
 n = n / 10;
 }
 if (rev == temp)
 printf("Palindrome Number");
 else
 printf("Not Palindrome Number");
 return 0;
```

}



6. Write a C program to find the GCD of two numbers.

```
#include <stdio.h>
int main() {
 int a, b, i, gcd;
 printf("Enter two numbers: ");
 scanf("%d %d", &a, &b);
 for (i = 1; i <= a && i <= b; i++) {
 if (a % i == 0 && b % i == 0)
 gcd = i;
 }
 printf("GCD = %d", gcd);
 return 0;
}
```

main.c

```c
//sheikh wasimuddin CSEIOT_63

#include <stdio.h>
int main() {
 int a, b, i, gcd;
 printf("Enter two numbers: ");
 scanf("%d %d", &a, &b);
 for (i = 1; i <= a && i <= b; i++) {
 if (a % i == 0 && b % i == 0)
 gcd = i;
 }
 printf("GCD = %d", gcd);
 return 0;
}
```

```
Enter two numbers: 10 50
GCD = 10

...Program finished with exit code 0
Press ENTER to exit console.
```

7. Assume that we want to create an array to store 5 integers. Since the size of int is 4 bytes, we need 5 * 4 bytes = 20 bytes of memory.

```c
#include <stdio.h>
int main() {
 int arr[5];
 printf("Size of int = %lu bytes\n", sizeof(int));
 printf("Size of array = %lu bytes\n", sizeof(arr));
 return 0;
}
```

main.c

```c
//sheikh wasimuddin CSEIOT_63
#include <stdio.h>
int main() {
 int arr[5];
 printf("Size of int = %lu bytes\n", sizeof(int));
 printf("Size of array = %lu bytes\n", sizeof(arr));
 return 0;
}
```

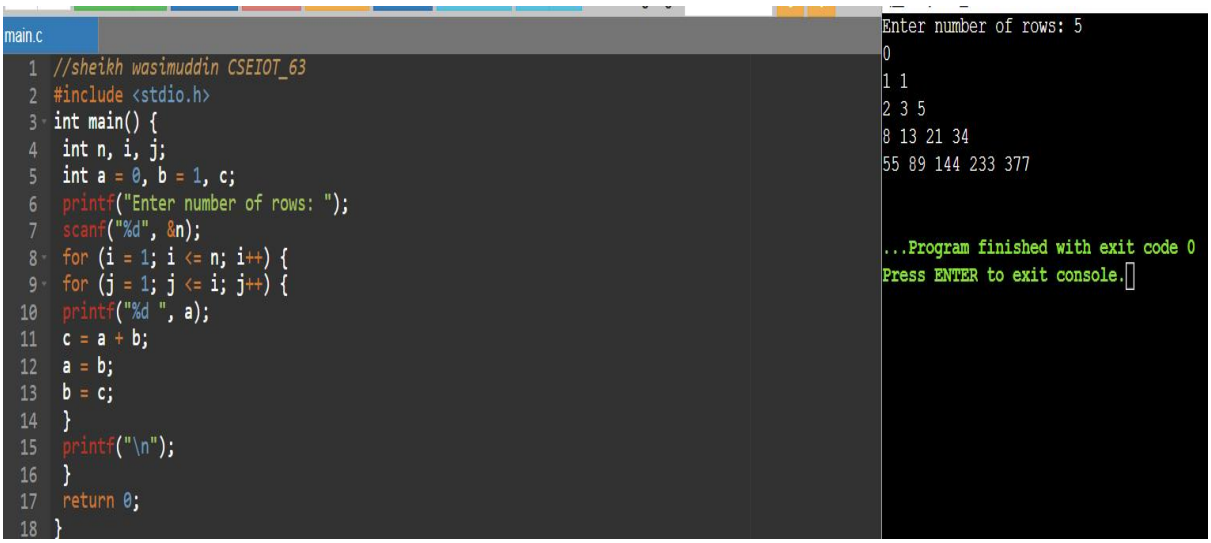
```
Size of int = 4 bytes
Size of array = 20 bytes

...Program finished with exit code 0
Press ENTER to exit console.
```

8. C Program to generate Fibonacci Triangle.

```c
#include <stdio.h>
int main() {
```

```
int n, i, j;
int a = 0, b = 1, c;
printf("Enter number of rows: ");
scanf("%d", &n);
for (i = 1; i <= n; i++) {
for (j = 1; j <= i; j++) {
printf("%d ", a);
c = a + b;
a = b;
b = c;
}
printf("\n");
}
return 0;
}
```



9. Implement a linked list in C and perform operations like insertion, deletion, and traversal.

```
#include <stdio.h>
```

```
#include <stdlib.h>
struct Node {
 int data;
 struct Node *next;
};
struct Node *head = NULL;
void insert(int value) {
 struct Node *newNode = (struct Node *)malloc(sizeof(struct
Node));
 newNode->data = value;
 newNode->next = NULL;
 if (head == NULL)
 head = newNode;
 else {
 struct Node *temp = head;
 while (temp->next != NULL)
 temp = temp->next;
 temp->next = newNode;
 }
}
void deleteNode(int value) {
 struct Node *temp = head, *prev = NULL;
 if (temp != NULL && temp->data == value) {
 head = temp->next;
 free(temp);
 return;
 }
 while (temp != NULL && temp->data != value) {
```

```
 prev = temp;
 temp = temp->next;
 }
 if (temp == NULL) {
 printf("Value not found\n");
 return;
 }
 prev->next = temp->next;
 free(temp);
}
void display() {
 struct Node *temp = head;
 while (temp != NULL) {
 printf("%d -> ", temp->data);
 temp = temp->next;
 }
 printf("NULL\n");
}
int main() {
 insert(10);
 insert(20);
 insert(30);
 printf("Linked List: ");
 display();
 deleteNode(20);
 printf("After Deletion: ");
 display();
 return 0;
```

}

main.c

```c
//sheikh wasimuddin CSEIOT_63
#include <stdio.h>
#include <stdlib.h>
struct Node {
 int data;
 struct Node *next;
};
struct Node *head = NULL;
void insert(int value) {
 struct Node *newNode = (struct Node *)malloc(sizeof(struct Node));
 newNode->data = value;
 newNode->next = NULL;
 if (head == NULL)
 head = newNode;
 else {
 struct Node *temp = head;
 while (temp->next != NULL)
 temp = temp->next;
 temp->next = newNode;
```

```
Linked List: 10 -> 20 -> 30 -> NULL
After Deletion: 10 -> 30 -> NULL


...Program finished with exit code 0
Press ENTER to exit console.
```

10. Create a memory management module in C that simulates allocation and deallocation of memory blocks.

```c
#include <stdio.h>
#define SIZE 10
int memory[SIZE];
void display() {
 int i;
 printf("\nMemory Blocks:\n");
 for (i = 0; i < SIZE; i++) {
 if (memory[i] == 0)
 printf("Block %d : Free\n", i);
 else
 printf("Block %d : Allocated\n", i);
 }
}
void allocate(int block) {
 if (block >= 0 && block < SIZE) {
 if (memory[block] == 0) {
```

```
 memory[block] = 1;
 printf("Block %d Allocated Successfully\n", block);
 } else {
 printf("Block %d Already Allocated\n", block);
 }
 } else {
 printf("Invalid Block Number\n");
 }
}
void deallocate(int block) {
 if (block >= 0 && block < SIZE) {
 if (memory[block] == 1) {
 memory[block] = 0;
 printf("Block %d Freed Successfully\n", block);
 } else {
 printf("Block %d Already Free\n", block);
 }
 } else {
 printf("Invalid Block Number\n");
 }
}
int main() {
 int choice, block;
 while (1) {
 printf("\n1. Allocate Memory");
 printf("\n2. Deallocate Memory");
 printf("\n3. Display Memory");
 printf("\n4. Exit");
```

```
printf("\nEnter choice: ");
scanf("%d", &choice);
if (choice == 1) {
printf("Enter block number (0-9): ");
scanf("%d", &block);
allocate(block);
}
else if (choice == 2) {
printf("Enter block number (0-9): ");
scanf("%d", &block);
deallocate(block);
}
else if (choice == 3) {
display();
}
else if (choice == 4) {
break;
}
else {
printf("Invalid Choice\n");
}
}
return 0;
}
```

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main.c

```c
//sheikh wasimuddin CSEIOT_63
#include <stdio.h>
#define SIZE 10
int memory[SIZE];
void display() {
 int i;
 printf("\nMemory Blocks:\n");
 for (i = 0; i < SIZE; i++) {
 if (memory[i] == 0)
 printf("Block %d : Free\n", i);
 else
 printf("Block %d : Allocated\n", i);
 }
}
void allocate(int block) {
 if (block >= 0 && block < SIZE) {
 if (memory[block] == 0) {
 memory[block] = 1;
 printf("Block %d Allocated Successfully\n", block);
 } else {
 printf("Block %d Already Allocated\n", block);
 }
 } else {
 printf("Invalid Block Number\n");
 }
}
void deallocate(int block) {
 if (block >= 0 && block < SIZE) {
 if (memory[block] == 1) {
 memory[block] = 0;
 printf("Block %d Freed Successfully\n", block);
 } else {
 printf("Block %d Already Free\n", block);
 }
 } else {
```

```
1. Allocate Memory
2. Deallocate Memory
3. Display Memory
4. Exit
Enter choice: 1
Enter block number (0-9): 5
Block 5 Allocated Successfully

1. Allocate Memory
2. Deallocate Memory
3. Display Memory
4. Exit
Enter choice: 1
Enter block number (0-9): 7
Block 7 Allocated Successfully

1. Allocate Memory
2. Deallocate Memory
3. Display Memory
4. Exit
Enter choice: 3

Memory Blocks:
Block 0 : Free
Block 1 : Free
Block 2 : Free
Block 3 : Free
Block 4 : Free
Block 5 : Allocated
Block 6 : Free
Block 7 : Allocated
Block 8 : Free
Block 9 : Free
```

main.c

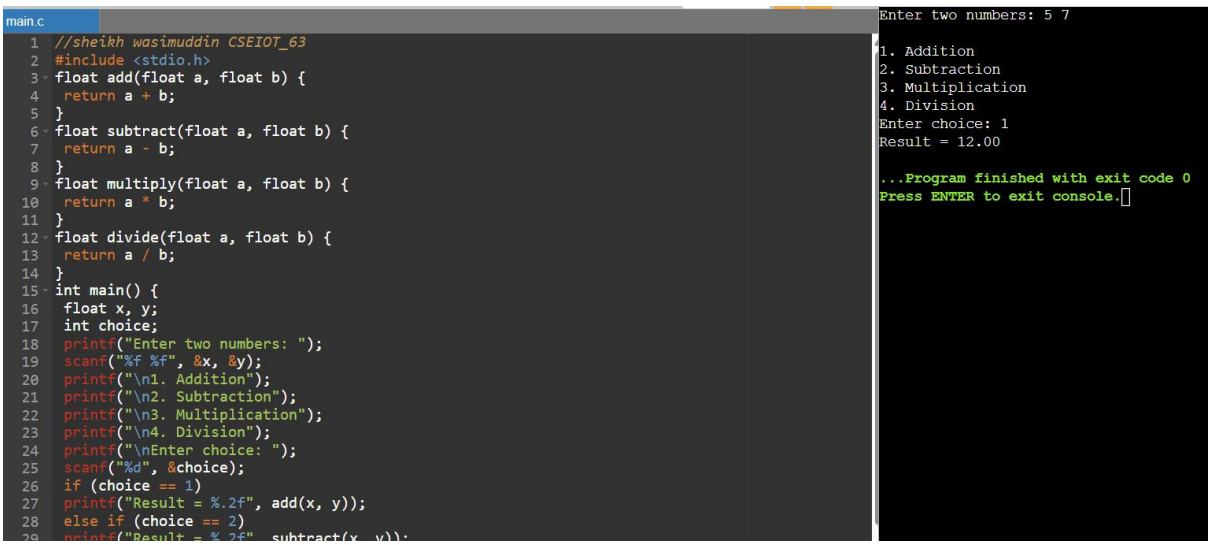
```c
//sheikh wasimuddin CSEIOT_63
#include <stdio.h>
#define SIZE 10
int memory[SIZE];
void display() {
 int i;
 printf("\nMemory Blocks:\n");
 for (i = 0; i < SIZE; i++) {
 if (memory[i] == 0)
 printf("Block %d : Free\n", i);
 else
 printf("Block %d : Allocated\n", i);
 }
}
void allocate(int block) {
 if (block >= 0 && block < SIZE) {
 if (memory[block] == 0) {
 memory[block] = 1;
 printf("Block %d Allocated Successfully\n", block);
 } else {
 printf("Block %d Already Allocated\n", block);
 }
 } else {
 printf("Invalid Block Number\n");
 }
}
void deallocate(int block) {
 if (block >= 0 && block < SIZE) {
 if (memory[block] == 1) {
 memory[block] = 0;
 printf("Block %d Freed Successfully\n", block);
 } else {
 printf("Block %d Already Free\n", block);
 }
 } else {
```

```
1. Allocate Memory
2. Deallocate Memory
3. Display Memory
4. Exit
Enter choice: 2
Enter block number (0-9): 5
Block 5 Freed Successfully

1. Allocate Memory
2. Deallocate Memory
3. Display Memory
4. Exit
Enter choice: 3

Memory Blocks:
Block 0 : Free
Block 1 : Free
Block 2 : Free
Block 3 : Free
Block 4 : Free
Block 5 : Free
Block 6 : Free
Block 7 : Allocated
Block 8 : Free
Block 9 : Free

1. Allocate Memory
2. Deallocate Memory
3. Display Memory
4. Exit
Enter choice: 4


...Program finished with exit code 0
```

11. Design a modular calculator application in C with functionalities like addition, subtraction, multiplication, and division.

```
#include <stdio.h>
float add(float a, float b) {
 return a + b;
}
float subtract(float a, float b) {
 return a - b;
}
float multiply(float a, float b) {
 return a * b;
}
float divide(float a, float b) {
 return a / b;
}
int main() {
 float x, y;
 int choice;
 printf("Enter two numbers: ");
 scanf("%f %f", &x, &y);
 printf("\n1. Addition");
 printf("\n2. Subtraction");
 printf("\n3. Multiplication");
 printf("\n4. Division");
 printf("\nEnter choice: ");
 scanf("%d", &choice);
 if (choice == 1)
```

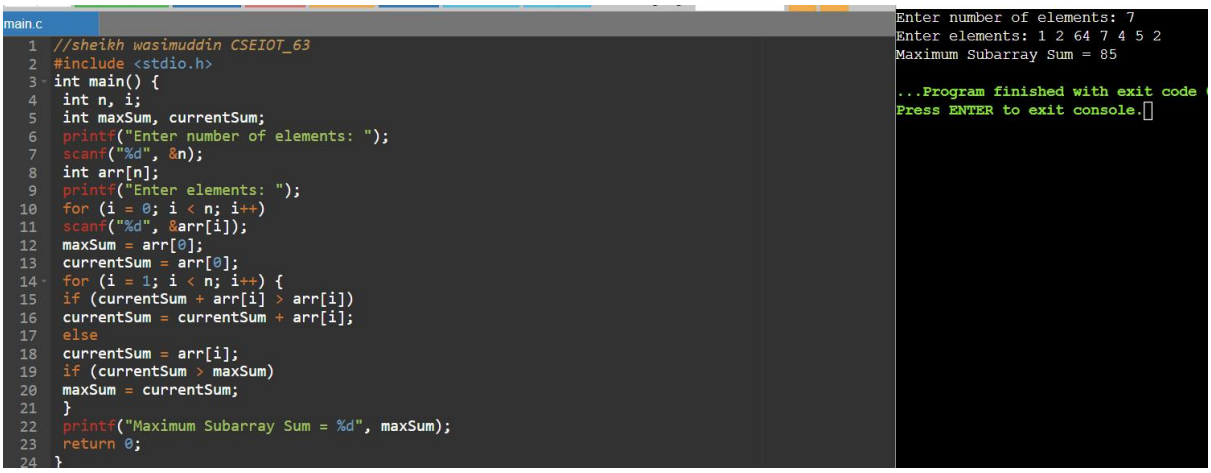
```
printf("Result = %.2f", add(x, y));
else if (choice == 2)
printf("Result = %.2f", subtract(x, y));
else if (choice == 3)
printf("Result = %.2f", multiply(x, y));
else if (choice == 4) {
if (y != 0)
printf("Result = %.2f", divide(x, y));
else
printf("Division not possible");
}
else
printf("Invalid choice");
return 0;
}
```



12. C Program on Kadane's Algorithm

```
#include <stdio.h>
int main() {
 int n, i;
```

```
int maxSum, currentSum;
printf("Enter number of elements: ");
scanf("%d", &n);
int arr[n];
printf("Enter elements: ");
for (i = 0; i < n; i++)
scanf("%d", &arr[i]);
maxSum = arr[0];
currentSum = arr[0];
for (i = 1; i < n; i++) {
if (currentSum + arr[i] > arr[i])
currentSum = currentSum + arr[i];
else
currentSum = arr[i];
if (currentSum > maxSum)
maxSum = currentSum;
}
printf("Maximum Subarray Sum = %d", maxSum);
return 0;
}
```



13. Check whether a number is a power of 2.

```
#include <stdio.h>
int main() {
 int n;
 printf("Enter number: ");
 scanf("%d", &n);
 if (n > 0 && (n & (n - 1)) == 0)
 printf("Power of 2");
 else
 printf("Not Power of 2");
 return 0;
}
```

14. C Program to Find the Maximum Value in an Array.

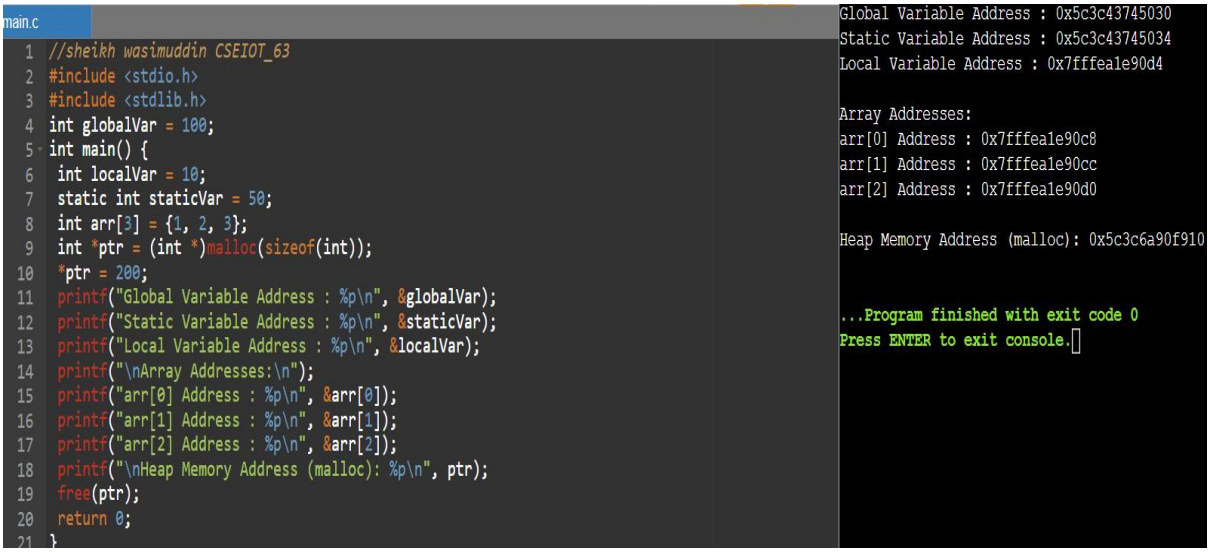
```
#include <stdio.h>
int main() {
 int n, i, max;
 printf("Enter number of elements: ");
 scanf("%d", &n);
 int arr[n];
 printf("Enter elements: ");
 for (i = 0; i < n; i++)
```

```
 scanf("%d", &arr[i]);
 max = arr[0];
 for (i = 1; i < n; i++) {
 if (arr[i] > max)
 max = arr[i];
 }
 printf("Maximum value = %d", max);
 return 0;
}
```

15. Write a C program to verify the Memory Layout.

```
#include <stdio.h>
#include <stdlib.h>
int globalVar = 100;
int main() {
 int localVar = 10;
 static int staticVar = 50;
 int arr[3] = {1, 2, 3};
 int *ptr = (int *)malloc(sizeof(int));
 *ptr = 200;
 printf("Global Variable Address : %p\n", &globalVar);
 printf("Static Variable Address : %p\n", &staticVar);
 printf("Local Variable Address : %p\n", &localVar);
```

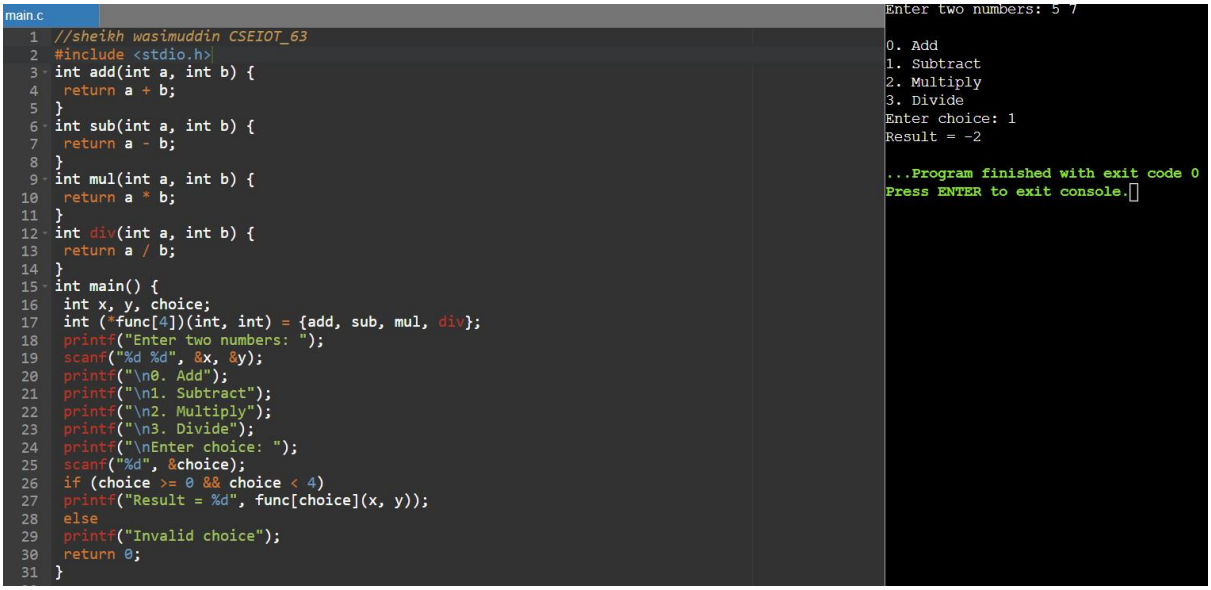
```
 printf("\nArray Addresses:\n");
 printf("arr[0] Address : %p\n", &arr[0]);
 printf("arr[1] Address : %p\n", &arr[1]);
 printf("arr[2] Address : %p\n", &arr[2]);
 printf("\nHeap Memory Address (malloc): %p\n", ptr);
 free(ptr);
 return 0;
}
```



16. Write a C program for function pointers in arrays to implement a set of functions dynamically.

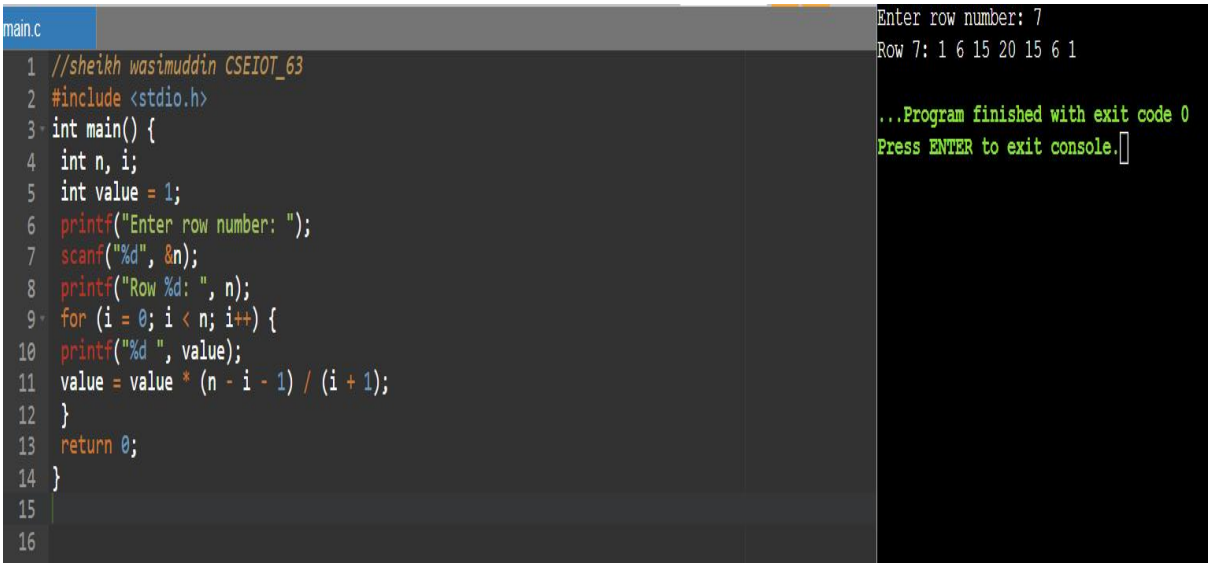
```
#include <stdio.h>
int add(int a, int b) {
 return a + b;
}
int sub(int a, int b) {
 return a - b;
}
int mul(int a, int b) {
```

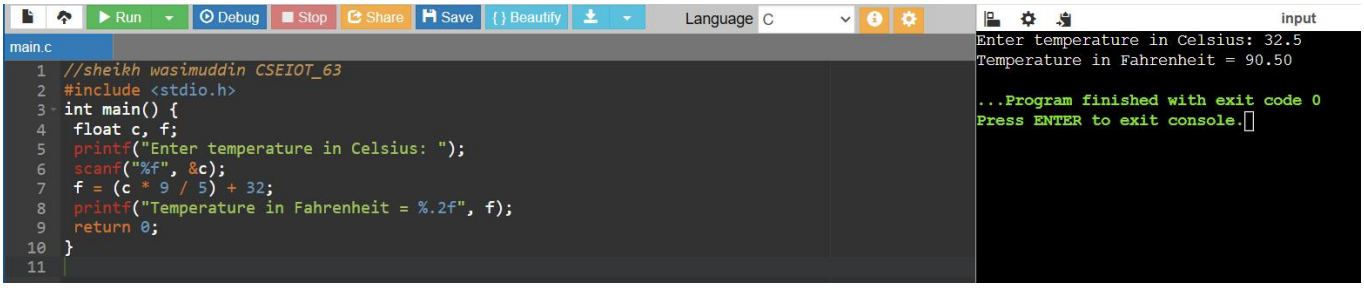
```
 return a * b;
}
int div(int a, int b) {
 return a / b;
}
int main() {
 int x, y, choice;
 int (*func[4])(int, int) = {add, sub, mul, div};
 printf("Enter two numbers: ");
 scanf("%d %d", &x, &y);
 printf("\n0. Add");
 printf("\n1. Subtract");
 printf("\n2. Multiply");
 printf("\n3. Divide");
 printf("\nEnter choice: ");
 scanf("%d", &choice);
 if (choice >= 0 && choice < 4)
 printf("Result = %d", func[choice](x, y));
 else
 printf("Invalid choice");
 return 0;
}
```



17. Write a Program to return the 5th row (or you can say nth row) of Pascal's triangle.

```
#include <stdio.h>
int main() {
 int n, i;
 int value = 1;
 printf("Enter row number: ");
 scanf("%d", &n);
 printf("Row %d: ", n);
 for (i = 0; i < n; i++) {
 printf("%d ", value);
 value = value * (n - i - 1) / (i + 1);
 }
 return 0;
}
```



18. Write a C program to convert degree Celsius into Fahrenheit.

```
#include <stdio.h>
int main() {
 float c, f;
 printf("Enter temperature in Celsius: ");
 scanf("%f", &c);
 f = (c * 9 / 5) + 32;
 printf("Temperature in Fahrenheit = %.2f", f);
 return 0;
}
```